Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Development of a Chiroptical-Based In-Line Biomanufacturing Module for In Situ, Biocompatible Process Monitoring			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio·medical		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Bio-signal Measurement / Diagnostic Devices	Fine Chemicals	Nano-Application Technologies
Concept of Proposed Technology	Biomanufacturing processes involve complex, dynamic biomolecular systems in which product quality and process performance are governed not only by concentration, but also by molecular structure, higher-order conformation, and chirality-dependent states. However, most existing in-line and at-line analytical techniques rely on intensity- or concentration-based readouts, such as absorbance, fluorescence, or refractive index changes. The proposed chiroptical approach addresses this fundamental gap by enabling direct optical access to molecular asymmetry, structural organization, and chirality-sensitive interactions, which are intrinsically difficult—and in many cases impractical—to resolve using conventional analytical modalities. Conventional planar chiral plasmonic architectures often lack the mechanical stability and biocompatibility required for long-term, continuous in-line bioprocessing, motivating the development of a composite-based chiral plasmonic platform. The proposed technology is an integrated chiral plasmonic composite platform designed for in-line and in situ monitoring in biomanufacturing environments, where biocompatibility and long-term operational stability constitute fundamental constraints. To address these requirements, chiral plasmonic nanostructures are embedded within biocompatible composite matrices, thereby enabling stable, long-term operation under complex biological processing conditions while generating chirality- and structure-sensitive chiroptical responses.[1–5] Localized superchiral field enhancement generated by chiral plasmonic nanostructures enables access to structure- and chirality-sensitive optical information that is orthogonal to conventional intensity-based sensing, while engineered nano–bio interfaces and suppression of non-specific interactions ensure reliable performance under realistic process conditions.[6–10] Coupled with data-			

	driven chiroptical signal analysis, the platform delivers robust and interpretable outputs, establishing a scalable technological foundation for real-time process monitoring and quality control (QC).
Rationale for Joint R&D Project	Advancing chiral plasmonic sensing technologies beyond proof-of-concept toward TRL 4 and higher requires tightly integrated development across materials design, nano–bio interfaces engineering, and system-level implementation under realistic process conditions, capabilities that cannot be achieved within a single organization. This joint R&D project is therefore structured to bridge fundamental chiral plasmonic innovation with biomanufacturing-relevant, in-line monitoring system development, enabling translation from laboratory demonstrations to deployable process technologies. The research institute (RPI) provides core leadership in chiral plasmonic nanostructure design, superchiral field engineering, and structure-sensitive chiroptical signal generation[11–22] (Ji-Young Kim, Ph.D.), together with complementary expertise in biomolecular target selection and nano–bio interface validation across representative biomanufacturing systems[23–29] (Jonathan S. Dordick, Ph.D.), establishing the scientific foundation required for selective, structure-resolved sensing. In close collaboration, the Korean SME leads the co-development of in-line–compatible system architectures and data-driven optical signal processing strategies, translating complex chiroptical responses into reliable monitoring outputs under continuous biomanufacturing operation. This integrated partnership directly addresses key TRL bottlenecks, including system integration, operational stability, and signal interpretability, ensuring the industrial relevance and deployability required for practical in-line manufacturing monitoring.
Key Words	1. Biocompatible plasmonic composite platforms 2. Superchiral field–enhanced detection 3. Nano–bio interface engineering 4. In-line biomanufacturing monitoring

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	**Start of the Project**	Level (2-3)	**End of the Project**	Level (4-5)
R&D Objectives	Biomanufacturing is a rapidly expanding industrial sector encompassing pharmaceutical production, cell and gene therapies, vaccines, and food biotechnology, where real-time, in-line monitoring of product quality attributes is increasingly critical for ensuring safety, consistency, and batch-to-batch reproducibility.[30,31] Despite this need, many current workflows continue to rely on off-line analysis and post-process quality control (QC), resulting in delayed feedback and limited sensitivity to subtle structural, chemical, or compositional variations that can critically impact product performance.[32,33] The overarching objective of this R&D project is to establish a chiral plasmonic composite–based monitoring platform as a core enabling technology for in-line and in situ assessment of structure-, chemistry-, and chirality-sensitive quality attributes across a broad range of biomanufactured products. Targeted attributes include recombinant and therapeutic proteins, monoclonal antibodies and antibody fragments, higher-order protein structure and folding integrity, post-translational modifications such as glycosylation, secreted biomolecules in cell-based manufacturing, chirality-relevant metabolites and amino acid compositions, as well as chirality-critical small-molecule drugs and drug intermediates.[34–36] Beyond biomanufacturing, the proposed platform is inherently suited for cell culture–based monitoring, enabling non-destructive, real-time observation of molecular and metabolic responses in living cells. By providing access to structure- and chirality-sensitive optical information that is orthogonal to conventional analytical readouts, the platform supports fundamental studies of cellular response mechanisms, drug mode-of-action analysis, and phenotype-level monitoring in chemically			

	and biologically dynamic environments.[1,2,33,37,38] Through this dual applicability, the project positions chiral plasmonic sensing as both a foundational technology for next-generation biomanufacturing QC and a versatile analytical platform for cell-based basic and translational research.
R&D Target Specifications	The final target is to demonstrate an in-line (in situ) chiral plasmonic composite monitoring system that exhibits stable operation, low background interference, and data-driven signal interpretation under biomanufacturing-relevant laboratory conditions. More detailed specifications are followings: **1. Chiroptical Performance of Chiral Plasmonic Composite** This target focuses on achieving strong and stable chiroptical performance of the final chiral plasmonic composite designed for biomanufacturing environments. The composite will be optimized to generate enhanced chiroptical responses through superchiral field effects while maintaining optical stability during long-term operation under biologically relevant conditions. Chiroptical response (e.g., g-factor): ≥ 0.02 Superchiral field enhancement factor: ≥ 2 Long-term optical performance retention (≥ 80%, over multi-week laboratory testing) **2. Biocompatibility and Stability for In-Line Monitoring Systems** This target aims to ensure that the chiral plasmonic composite platform is biocompatible and physically stable for in-line (in situ) monitoring applications. The platform will be evaluated under biomanufacturing-relevant conditions to confirm the absence of cytotoxic effects and the preservation of mechanical and thermal stability during continuous operation. Cytotoxicity: No significant cytotoxicity (cell viability ≥ 95%) Mechanical stability: Maintained under continuous flow and handling conditions Thermal stability: Stable operation under 45 °C **3. Sensitivity and Selectivity Enhancement via Nano–Bio Interface Engineering** This target aims to improve sensitivity and selectivity through nano–bio interface engineering while reducing non-specific binding and cross talk in complex biomanufacturing environments. By minimizing interference from non-target species, structure-sensitive chiroptical responses can be preserved across multiple biomanufacturing targets. Detection sensitivity: Picomolar-level detection (LOD ~ picomolar level) Reduction of non-specific binding and cross talk (≤ 10%) Applicability across biomanufacturing targets (≥ 3 distinct target models) **4. Data-Driven Chiroptical Signal Analysis and Modeling** This target aims to enhance overall monitoring performance through data-driven analysis and modeling of chiroptical signals. AI and machine learning–based tools will be applied to interpret complex signals, suppress background interference, and improve robustness against time-dependent drift in long-term monitoring scenarios. Background signal suppression efficiency (≥ 80%) Signal classification or interpretation accuracy (≥ 90%) Robust performance under time-dependent drift and complex background conditions
R&D Plan	**Aim 1. Initiation of RPI-Korean SME collaboration** (0–6 months) This aim focuses on initiating the collaboration between the PIs at RPI and a Korean SME. We will define the target biomanufacturing systems and select relevant target biomolecules, and will establish design strategies for chiral plasmonic nano–bio interfaces and biocompatible chiral plasmonic–hydrogel composites to enable sensitive and structure-specific detection. In parallel, a concrete plan for securing and managing intellectual property arising from these developments will be formulated. These efforts will culminate in the joint preparation of a full proposal by RPI team and the Korean SME. ***1.1. Selection of target biomanufacturing systems and relevant biomolecule candidates*** *(Kim-Dordick-Korean SME)* Target biomanufacturing systems will be defined to represent key stages of bioprocess

development, monitoring, and quality control (QC), including mammalian cell culture and bioreactor platforms, viral vector and lentiviral manufacturing, and downstream purification and QC processes. For each system, relevant biomolecules (e.g., DNA/RNA, proteins, exosomes, and lentiviral vectors) will be identified based on their impact on process performance and product quality. These systems and biomolecules will be matched to select a final set of **representative system–biomolecule pairs** for sensor design and validation.

1.2. Design of chiral plasmonic nano–bio interfaces and their composites *(Kim-Korean SME)*
This project establishes design strategies for chiral plasmonic nano–bio interfaces and biocompatible composite platforms for biomanufacturing applications through close collaboration between RPI and the Korean SME. Chiral plasmonic nanostructures based on gold and silver will be synthesized using proprietary techniques developed in Kim's group, enabling tunable optical activity across the visible to near-infrared (NIR) range. Nano–bio interfaces will be functionalized with target-specific chemistries, including peptides, proteins, antibodies, or aptamers, to achieve selective biomolecular recognition. These chiral plasmonic nanostructures will be integrated into representative biocompatible matrices such as hydrogels (e.g., poly(ethylene glycol) diacrylate (PEGDA), cellulose, and alginate) and biocompatible polymers (e.g., poly(vinyl alcohol) (PVA) and poly(lactic-co-glycolic acid) (PLGA)) with flexible adaptation to diverse biological and cell culture environments.

Aim 2 - Foundational Engineering of Chiral Plasmonic and Biocompatible Interfaces (0–12 months)
Aim 2 focuses on establishing the core technical foundations required for chiral plasmonic biosensing. This aim develops both high-performance nano–bio interfaces and biocompatible composite materials that enable sensitive and selective chiroptical signal generation in bio-target detection.

2.1. Chiral plasmonic nano–bio interface engineering *(Kim-Dordick)*
Chiral plasmonic nano–bio interfaces will be designed to enhance biosensing sensitivity and selectivity through controlled plasmon–biomolecule interactions.

2.2. Biocompatible chiral plasmonic–matrix composite development *(Kim-Korean SME)*
This aim develops biocompatible composite platforms that embed chiral plasmonic structures within matrices suitable for biological and cell culture environments, including representative hydrogels and biocompatible polymer systems, to ensure compatibility and stability under bioprocessing-relevant conditions.

Aim 3 - Integration and Generalization of Biocompatible Chiral Plasmonic Biosensing Platforms (7–24 months)
We will integrate the individual components developed in Aim 2 into functional biosensing platforms and systemically expands their applicability across diverse biomanufacturing targets and conditions. The emphasis is on generalizing the sensing strategy beyond a single target or environment.

3.1. Integrated biocompatible chiral plasmonic biosensing platforms (Kim-Dordick-Korean SME)
This aim integrates nano–bio interfaces and biocompatible composites into a unified chiral plasmonic biosensing platform capable of reliable operation in biological media.

3.2. Target-tailored nano–bio interface design for diverse applications (Kim-Dordick)
This aim develops target-specific nano–bio interface design strategies that preserve sensitivity and selectivity across different biomanufacturing processes, thereby expanding the generality of chiral plasmonic sensors.

Aim 4 - Preparation for In-Line / In Situ Monitoring through Long-Term Stability and Model Biomanufacturing Studies (19–30 months)
We will prepare the sensing platform for future in-line and in situ deployment by addressing long-term operational stability under complex biomanufacturing-like environments. This aim focuses on mitigating performance degradation caused by non-specific adsorption and environmental interference.

4.1. Long-term stability optimization in complex bioenvironments (Kim-Dordick-Korean SME)

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	This aim minimizes cross talk and performance loss arising from non-specific adsorption and surface fouling while maintaining sensitivity and selectivity during prolonged operation. ***4.2. Laboratory-scale validation using model biomanufacturing systems*** (Kim-Dordick) This aim evaluates sensor performance in laboratory-scale, biomanufacturing-relevant model systems, allowing assessment of signal stability, selectivity retention, and robustness prior to in-line implementation. **Aim 5 - Optimization of Chiral Plasmonic Monitoring for Long-Term In-Line Applications** (25–36 months) Aim 5 focuses on optimizing the chiral plasmonic monitoring system as a whole, transforming it from a stable sensor into a robust monitoring tool suitable for long-term in-line applications. ***5.1. Data-driven optimization of chiroptical monitoring*** (Kim-Dordick-Korean SME) This aim applies AI- and machine learning–based optical signal models to interpret complex chiroptical data and ensure reliable long-term monitoring performance. **5.2. System-level optimization for reliable in-line monitoring** (Kim-Dordick-Korean SME) This aim optimizes sensing configurations, measurement strategies, and operational parameters to improve robustness, reproducibility, and readiness for in-line deployment.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	As illustrated in the figure (below), the research institute assumes primary responsibility for the scientific foundation and validation of the proposed technology. **Figure.** Bibliography, expertise, and selected outstanding research outputs of the principal investigators(PI) at the research institute ***Ji-Young Kim, Ph.D.*** is responsible for the design and optimization of chiral plasmonic nanostructures,[14,15,17–19,21,22] superchiral field generation,[14,17] and structure-sensitive chiroptical sensing principles.[14–16,18,20] ***Jonathan S. Dordick, Ph.D.*** is responsible for biointerface engineering, selection of biomanufacturing-relevant biomolecular targets,[25,27,28] and ensuring biological relevance and generality across biomanufacturing processes.[24,26,29] Together, the two PIs are responsible for defining sensing mechanisms, performance benchmarks, and laboratory-scale validation strategies that underpin the chiral plasmonic in-line monitoring platform, while working closely with the Korean SME on composite integration and signal interpretation.

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Expected R&R of Korean SME Partner	The Korean SME partner will play a key role in translating the core sensing principles developed by the research institute into practical, in-line–compatible monitoring technologies. The SME's responsibilities focus on system integration, biocompatible composite development, and implementation strategies required for operation under realistic biomanufacturing conditions. First, the Korean SME will co-develop biocompatible composite architectures in close collaboration with the research institute, integrating chiral plasmonic nanostructures into hydrogel- and polymer-based matrices to achieve long-term stability, mechanical robustness, and compatibility with cell culture and bioprocessing environments. Second, the SME will lead the development of in-line–compatible sensing configurations and measurement workflows, including platforms suitable for continuous or semi-continuous operation and integration with flow-based environments, **with microfluidic implementations explored as an optional configuration when appropriate**. As part of this effort, the SME will work jointly with the research institute to implement signal handling and processing strategies that ensure stable, reliable readout under dynamic process conditions. Through these activities, the Korean SME partner will support the maturation of the chiral plasmonic composite platform beyond laboratory-scale validation, enabling system integration and operational readiness consistent with **TRL 4 and higher** for in-line biomanufacturing monitoring.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This project is expected to generate significant scientific and socio-economic value by establishing chiral plasmonic composite–based in-line monitoring as a core enabling technology for next-generation biomanufacturing. From an academic perspective, the project will advance fundamental understanding of superchiral field–mediated light–matter interactions at nano–bio interfaces, particularly under complex and dynamic biological environments relevant to real-world bioprocessing. By systematically linking chiroptical signal characteristics to biomolecular structure, state, and interaction dynamics, this work will contribute new conceptual and methodological frameworks for structure-sensitive biosensing and chiroptical analytics. The proposed technology is designed for adaptable integration across diverse biomanufacturing processes and biomolecular targets, supporting applications in protein therapeutics, cell and gene therapies, vaccines, and cell culture–based research. From an industrial perspective, the project directly addresses fundamental limitations of current biomanufacturing QC, which remains heavily reliant on labor-intensive, expertise-driven, and off-line analytical workflows. By enabling real-time, structure-sensitive in-line monitoring, the technology has the potential to reduce dependence on highly trained personnel, improve process robustness and reproducibility, and accelerate scale-up and technology transfer. These capabilities are expected to generate added value across pharmaceutical manufacturing, cell and gene therapies, vaccine production, and food biomanufacturing, while also laying the groundwork for downstream commercialization and workforce transformation. Importantly, the project supports a globally scalable value-creation model led by a Korean SME, enabling deployment of modular sensing and analytics solutions to international biomanufacturing enterprises, thereby strengthening global competitiveness and commercialization potential. **Quantitative performance, dissemination, and value-creation targets include:** High impact paper publications related to this work: ≥ 3 Q1 journal publications, ≥ 1 publication within the top 10% journals in the field Patent applications arising from this proposal: ≥ 1 patent filing in Korea and/or the United States

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization